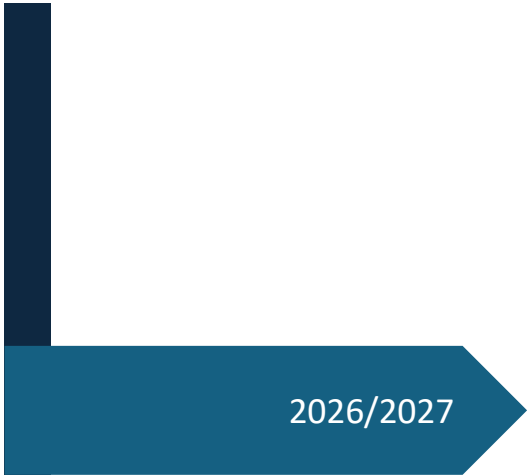
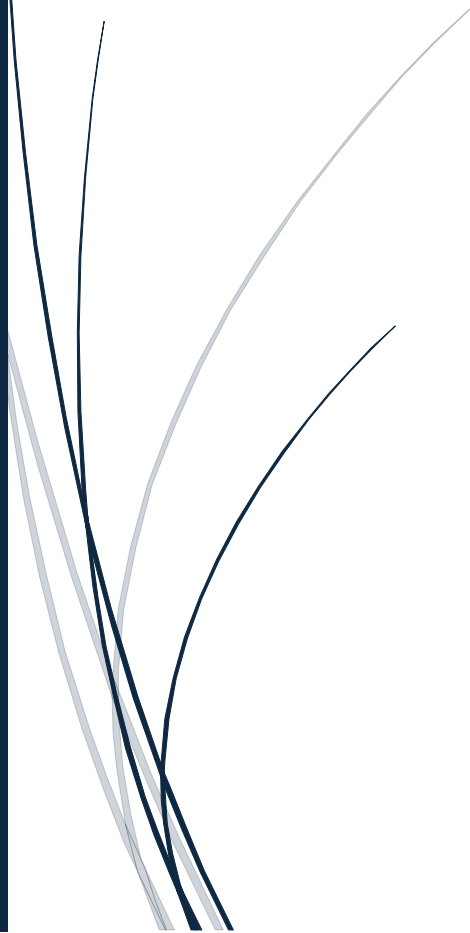


A dark blue vertical bar on the left side of the page. A blue arrow points to the right from the bar, containing the text '2026/2027'.

2026/2027

Instalación, configuración y documentación del entorno de desarrollo y del entorno de explotación

Several thin, curved lines in dark blue and light grey that sweep upwards from the bottom left corner of the page.

MOISÉS ALBERTO DOMÍNGUEZ CRUZ
IES LOS SAUCES – DAW2

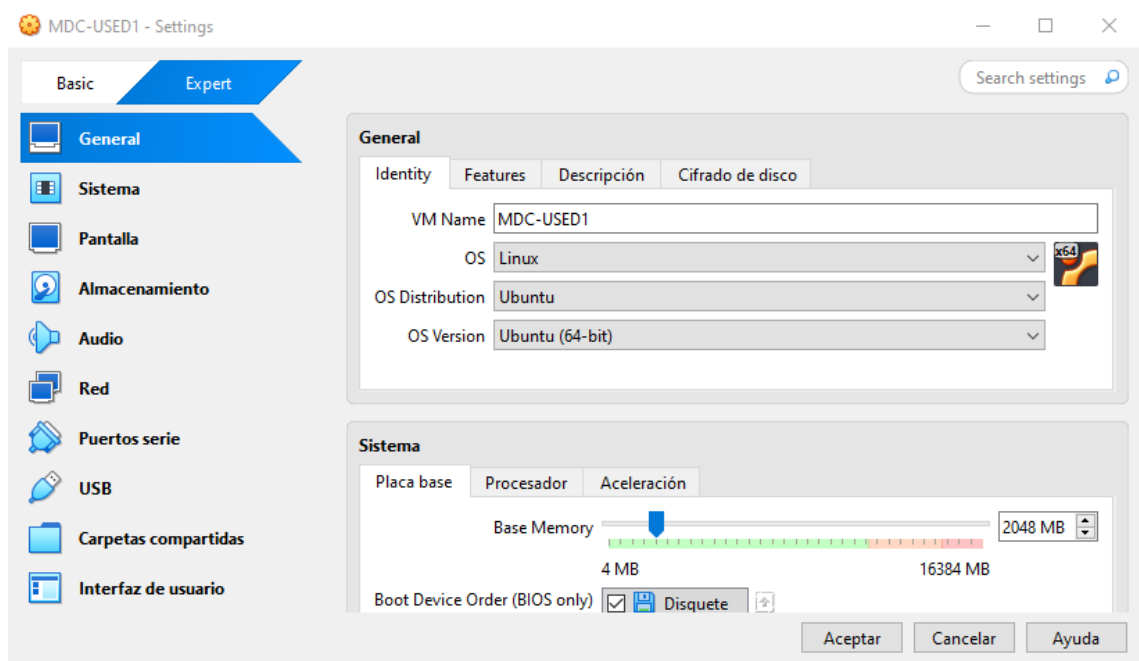
Configuración MDC-USED1

1. USED – Ubuntu Server.....	2
1.1 Configuración inicial	2
1.2 Apache	6
1.2 PHP.....	9

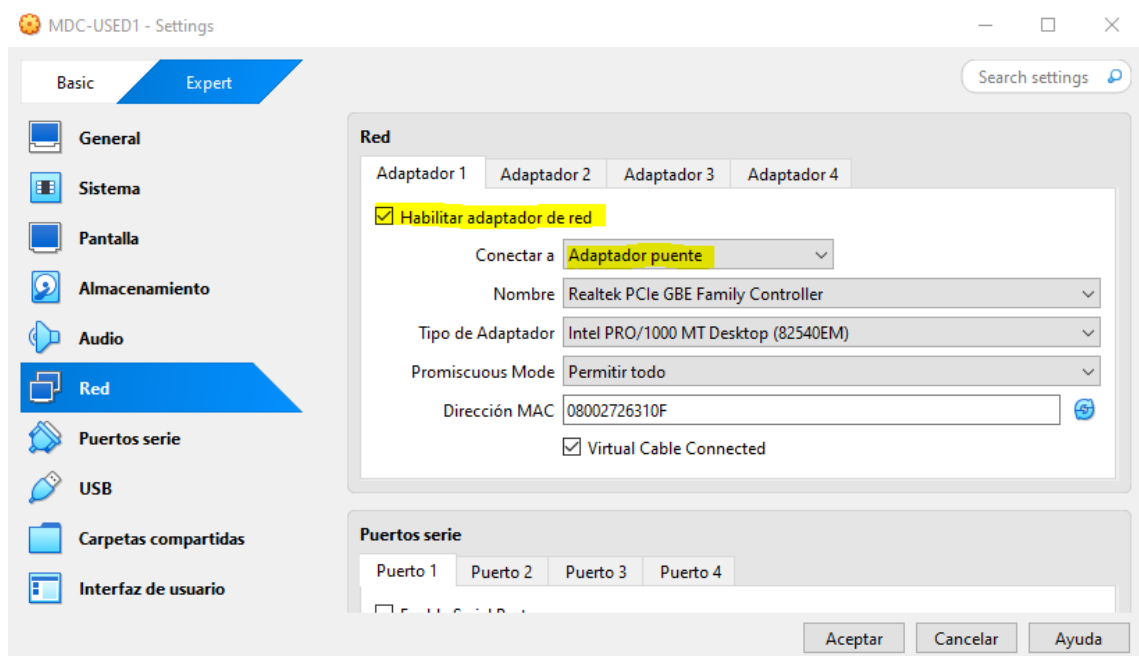
1. USED – Ubuntu Server

1.1 Configuración inicial

General → XXX-USED1 (XXX son las iniciales de la persona)



Red → Adaptador 1 (habilitamos y lo conectamos a Adaptador puente)



Comprobación de la red → ip a (comando para ver nuestra configuración de red)

```
miadmin@mdc-uslimpia: ~  
miadmin@mdc-uslimpia:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000  
    link/ether 00:00:00:00:00:00 brd ff:ff:ff:ff:ff:ff  
    altname enx08002726310f  
    inet 10.199.9.184/22 brd 10.199.11.255 scope global enp0s3  
        valid_lft forever preferred_lft forever  
    inet6 fe80::a00:27ff:fe26:310f/64 scope link proto kernel_l1  
        valid_lft forever preferred_lft forever  
miadmin@mdc-uslimpia:~$
```

Si tenemos que configurar nuestra red → cd /etc/netplan y hacemos un ls

Antes hacemos una copia de la configuración con el comando → sudo cp 00-installer-config.yaml 00-installer-config.yaml.backup

```
miadmin@mdc-uslimpia: /etc/netplan  
miadmin@mdc-uslimpia:~$ cd /etc/netplan  
miadmin@mdc-uslimpia:/etc/netplan$ ls  
00-installer-config.yaml 00-installer.config.backup  
miadmin@mdc-uslimpia:/etc/netplan$
```

Editar la configuración (Añadir IP 10.199.11.203/22, Gateway 10.199.8.1 y DNS 10.151.123.21, 10.151.126.21) → sudo nano 00-installer-config.yaml

```
miadmin@mdc-uslimpia: /etc/netplan  
GNU nano 8.7.1  
# This is the network config written by 'subiquity'  
network:  
  version: 2  
  ethernet:  
    enp0s3:  
      dhcp4: false  
      addresses:  
        - 10.199.9.184/22  
      routes:  
        - to: default  
          via: 10.199.8.1  
      nameservers:  
        addresses: [10.151.123.21, 10.151.126.21]  
        search: [educa.jcyl.es]
```

Aplicamos la configuración → sudo netplan apply

Cambiar nombre de la maquina → `sudo hostnamectl set-hostname nuevo-nombre`

```
miadmin@mdc-uslimpia: /  
miadmin@mdc-uslimpia:/$ sudo hostnamectl set-hostname mdc-used1  
miadmin@mdc-uslimpia:/$
```

También cambiamos en hosts → `sudo nano /etc/hosts`

```
miadmin@mdc-uslimpia: /  
GNU nano 8.7.1 /etc/hosts  
127.0.0.1 localhost  
127.0.1.1 mdc-used1  
  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters
```

Comprobamos el nombre → `hostname`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ hostname  
mdc-used1  
miadmin@mdc-used1:~$
```

Actualizar el sistema operativo → `sudo apt update && sudo apt upgrade -y --fix-missing`

```
miadmin@mdc-used1:~$ sudo apt update && sudo apt upgrade -y --fix-missing  
[sudo: authenticate] Password:  
Obj:1 http://archive.ubuntu.com/ubuntu resolute InRelease  
Des:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]  
Des:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]  
Des:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [550 kB]  
Ign:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease  
Des:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]  
Ign:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages  
Des:6 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [129 kB]  
Des:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46,7 kB]  
Err:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components  
La suma de comprobación difiere  
Hashes of expected file:  
- Filesize:46692 [weak]  
- SHA256:606c5dd880af82f5ebc7655f263bc24ac62b7b5f1cc03053b2b011289c746caa  
- SHA1:5e24539ba03c849fbb0afc9ad3ff4b571de93c10 [weak]  
- MD5Sum:9b3ba26dc1c5ad84df01efbabe45287f [weak]  
Hashes of received file:  
- SHA256:7728dff8fb17bd4e0510061bc1ce3c18d3cd41d663e7b7483a2c89027fd84d71  
- SHA1:dc25421e7aefff126a6dccc6f119fa784b9028f [weak]  
- MD5Sum:98874845fb7bd2415701b7f87f0d9e81 [weak]  
- Filesize:46692 [weak]  
Last modification reported: Sun, 04 Oct 2026 08:17:29 +0000  
Release file created at: Sun, 04 Oct 2026 11:12:38 +0000
```

Comprobar usuarios → `cat /etc/passwd | grep miadmin`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ cat /etc/passwd | grep miadmin  
miadmin:x:1000:1000:miadmin:/home/miadmin:/bin/bash  
miadmin2:x:1001:1001:,,,:/home/miadmin2:/bin/bash  
miadmin@mdc-used1:~$
```

Comprobar los discos y particiones → `lsblk`

```
miadmin@mdc-used1:~$ lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
loop0        7:0      0 44,7M  1 loop /snap/snapd/28254  
loop1        7:1      0 66,8M  1 loop /snap/core24/1643  
loop2        7:2      0  1,5M  1 loop /snap/hwctl/123  
loop3        7:3      0  1,5M  1 loop /snap/hwctl/131  
loop4        7:4      0 50,1M  1 loop /snap/snapd/27710  
sda          8:0      0 500G   0 disk  
├─sda1       8:1      0    1M   0 part  
├─sda2       8:2      0 150G   0 part /  
├─sda3       8:3      0    4G   0 part [SWAP]  
└─sda4       8:4      0 346G   0 part /var  
sr0         11:0     1 1024M  1 rom
```

1.2 Apache

Instalación del servicio Apache → `sudo apt install apache2`

```
miadmin@mdc-used1:~$ sudo apt install apache2
Instalando:
  apache2

Instalando dependencias:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64  liblua5.4-0  ssl-cert

Paquetes sugeridos:
  apache2-doc  apache2-suexec-pristine | apache2-suexec-custom  www-browser
```

Comprobamos el servicio → `sudo systemctl status apache2`

```
miadmin@mdc-used1:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 08:20:23 UTC; 21s ago
     Invocation: 1a13c7969df847c7b2d678a5656a2b40
       Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 5205 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 55 (limit: 1891)
    Memory: 5.6M (peak: 5.6M)
       CPU: 173ms
    CGroup: /system.slice/apache2.service
            └─5205 /usr/sbin/apache2 -k start -DFOREGROUND
              └─5225 /usr/sbin/apache2 -k start -DFOREGROUND
                └─5226 /usr/sbin/apache2 -k start -DFOREGROUND

oct 05 08:20:23 mdc-used1 systemd[1]: Starting apache2.service - The Apache HTTP Server...
oct 05 08:20:23 mdc-used1 apache2[5205]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to
oct 05 08:20:23 mdc-used1 systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

Abrimos el puerto http (80) → `sudo ufw allow http`

```
miadmin@mdc-used1:~$ sudo ufw allow http
Skipping adding existing rule
Skipping adding existing rule (v6)
miadmin@mdc-used1:~$
```

Ya que estamos comprobamos los demás servicios

Ssh → `sudo systemctl status ssh`

```
miadmin@mdc-used1:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 08:07:40 UTC; 15min ago
     Invocation: 5dae769c8b34435c9ad4903de2bbb612
   TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Main PID: 1786 (sshd)
       Tasks: 1 (limit: 1891)
      Memory: 6.1M (peak: 9.1M)
         CPU: 127ms
    CGroup: /system.slice/ssh.service
            └─1786 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

oct 05 08:07:38 mdc-used1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
oct 05 08:07:40 mdc-used1 sshd[1786]: Server listening on 0.0.0.0 port 22.
oct 05 08:07:40 mdc-used1 sshd[1786]: Server listening on :: port 22.
oct 05 08:07:40 mdc-used1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
oct 05 08:07:47 mdc-used1 sshd-session[1788]: Accepted password for miadmin from 10.199.11.203 port 51223 ssh2
oct 05 08:07:47 mdc-used1 sshd-session[1788]: pam_unix(sshd:session): session opened for user miadmin(uid=1000) by miadmin(uid=0)
miadmin@mdc-used1:~$
```

Firewall local → sudo systemctl status ufw

```
miadmin@mdc-used1:~$ sudo systemctl status ufw
● ufw.service - Uncomplicated firewall
   Loaded: loaded (/usr/lib/systemd/system/ufw.service; enabled; preset: enabled)
   Active: active (exited) since Mon 2026-10-05 08:07:17 UTC; 16min ago
  Invocation: 537750d11fda478397b1a8951cfd7a10
     Docs: man:ufw(8)
    Main PID: 1225 (code=exited, status=0/SUCCESS)
   Mem peak: 2.2M
      CPU: 210ms

oct 05 08:07:15 mdc-used1 systemd[1]: Starting ufw.service - Uncomplicated firewall...
oct 05 08:07:17 mdc-used1 systemd[1]: Finished ufw.service - Uncomplicated firewall.
miadmin@mdc-used1:~$
```

Antivirus → sudo systemctl status clamav-daemon

```
miadmin@mdc-used1:~$ sudo systemctl status clamav-daemon
● clamav-daemon.service - Clam AntiVirus userspace daemon
   Loaded: loaded (/usr/lib/systemd/system/clamav-daemon.service; enabled; preset: enabled)
   Drop-In: /etc/systemd/system/clamav-daemon.service.d
            └─extend.conf
   Active: active (running) since Mon 2026-10-05 08:07:20 UTC; 17min ago
  Invocation: 1dda3f9d976c4098aa0f3abc6ebe070d
 TriggeredBy: ● clamav-daemon.socket
     Docs: man:clamd(8)
           man:clamd.conf(5)
           https://docs.clamav.net/
    Main PID: 1575 (clamd)
       Tasks: 2 (limit: 1891)
      Memory: 738.9M (peak: 1G, swap: 219.4M, swap peak: 219.4M)
         CPU: 20.192s
    CGroup: /system.slice/clamav-daemon.service
            └─1575 /usr/sbin/clamd --foreground=true

oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> ELF support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> Mail files support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> OLE2 support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> PDF support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> SWF support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> HTML support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> XMLDOCS support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> HWP3 support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> OneNote support enabled.
oct 05 08:07:43 mdc-used1 clamd[1575]: Mon Oct 5 08:07:43 2026 -> Self checking every 3600 seconds.
miadmin@mdc-used1:~$
```

sudo systemctl status clamav-freshclam

```
miadmin@mdc-used1:~$ sudo systemctl status clamav-freshclam
● clamav-freshclam.service - ClamAV virus database updater
   Loaded: loaded (/usr/lib/systemd/system/clamav-freshclam.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 08:07:19 UTC; 18min ago
  Invocation: 702c486319ca4f489061c86e75d7b924
     Docs: man:freshclam(1)
           man:freshclam.conf(5)
           https://docs.clamav.net/
    Main PID: 1537 (freshclam)
       Tasks: 1 (limit: 1891)
      Memory: 16.2M (peak: 33.5M, swap: 256K, swap peak: 256K)
         CPU: 243ms
    CGroup: /system.slice/clamav-freshclam.service
            └─1537 /usr/bin/freshclam -d --foreground=true

oct 05 08:07:19 mdc-used1 systemd[1]: Started clamav-freshclam.service - ClamAV virus database updater.
oct 05 08:07:23 mdc-used1 freshclam[1537]: Mon Oct 5 08:07:23 2026 -> ClamAV update process started at Mon Oct 5 08:07:23 2026
oct 05 08:07:23 mdc-used1 freshclam[1537]: Mon Oct 5 08:07:23 2026 -> daily.cld database is up-to-date (version: 28144, sigs: 355713, f-level: 90, builder: svc.clamav-publisher)
oct 05 08:07:23 mdc-used1 freshclam[1537]: Mon Oct 5 08:07:23 2026 -> main.cvd database is up-to-date (version: 63, sigs: 3287027, f-level: 90, builder: tomjudge)
oct 05 08:07:23 mdc-used1 freshclam[1537]: Mon Oct 5 08:07:23 2026 -> bytecode.cvd database is up-to-date (version: 339, sigs: 80, f-level: 90, builder: nrandolp)
miadmin@mdc-used1:~$
```


Creamos cuenta operadorweb → El usuario operador web gestiona el contenido del servidor web con permisos limitados a `/var/www/html`. → `sudo useradd -M -d /var/www/html -g www-data -s /bin/bash operadorweb`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ sudo useradd -M -d /var/www/html -g www-data -s /bin/bash operadorweb  
miadmin@mdc-used1:~$
```

Establecemos una contraseña → `sudo passwd operadorweb`

```
miadmin@mdc-used1:~$ sudo passwd operadorweb  
New password:  
Retype new password:  
passwd: contraseña actualizada correctamente
```

Configurar grupo para operador web → `sudo chown -R operadorweb:www-data /var/www/html`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ sudo chown -R operadorweb:www-data /var/www/html  
miadmin@mdc-used1:~$
```

Establecer permisos → `sudo chmod -R 775 /var/www/html`

```
miadmin@mdc-used1:~$ sudo chmod -R 775 /var/www/html  
miadmin@mdc-used1:~$
```

Comprobamos operadorweb → `id operadorweb`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ id operadorweb  
uid=1002(operadorweb) gid=33(www-data) groups=33(www-data)  
miadmin@mdc-used1:~$
```

Verificar propiedad → `ls -la /var/www/`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ ls -la /var/www/  
total 12  
drwxr-xr-x  3 root      root      4096 Oct  5 08:20 .  
drwxr-xr-x 15 root      root      4096 Oct  5 08:20 ..  
drwxrwxr-x  2 operadorweb www-data 4096 Oct  5 08:20 html  
miadmin@mdc-used1:~$
```

Verificar permisos → `ls -ld /var/www/html`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ ls -ld /var/www/html  
drwxrwxr-x 2 operadorweb www-data 4096 Oct  5 08:20 /var/www/html  
miadmin@mdc-used1:~$
```

1.2 PHP

Instalar php8.5-fpm → `sudo apt install php8.5-fpm`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ sudo apt install php8.5-fpm  
Instalando:  
  php8.5-fpm  
  
Instalando dependencias:  
  libargon2-1 php-common php8.5-cli php8.5-common php8.5-readline  
  
Paquetes sugeridos:  
  php-pear  
  
Resumen:  
  Actualizando: 0, Instalando 6, Eliminando: 0, no actualizando: 0  
  Tamaño de la descarga: 7.141 kB  
  Espacio necesario: 36,8 MB / 146 GB disponible  
  
¿Continuar? [S/n] S  
Des:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libargon2-1 amd64 0~20190702+dfsg-5 [21,3 kB]  
Des:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 php-common all 2:99ubuntu1 [13,9 kB]  
Des:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 php8.5-common amd64 8.5.4-0ubuntu1.3 [816 kB]  
Ign:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 php8.5-common amd64 8.5.4-0ubuntu1.3  
Des:3 http://security.ubuntu.com/ubuntu resolute-updates/main amd64 php8.5-common amd64 8.5.4-0ubuntu1.3 [816 kB]  
15% [3 php8.5-common 719 kB/816 kB 88%] [Esperando las cabeceras]
```

Reiniciar el servicio php8.5-fpm → `sudo systemctl restart php8.5-fpm`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ sudo systemctl restart php8.5-fpm  
miadmin@mdc-used1:~$
```

Configurar apache2 con php8.5-fpm → `sudo a2enmod proxy_fcgi setenvif`

```
miadmin@mdc-used1: ~  
miadmin@mdc-used1:~$ sudo a2enmod proxy_fcgi setenvif  
Considering dependency proxy for proxy_fcgi:  
Enabling module proxy.  
Enabling module proxy_fcgi.  
Module setenvif already enabled  
To activate the new configuration, you need to run:  
  systemctl restart apache2  
miadmin@mdc-used1:~$
```

Activamos los dos servicios → `sudo a2enconf php8.5-fpm`

```
miadmin@mdc-used1:~$ sudo a2enconf php8.5-fpm
Enabling conf php8.5-fpm.
To activate the new configuration, you need to run:
  systemctl reload apache2
miadmin@mdc-used1:~$
```

Y ahora reiniciamos servicios → `sudo systemctl restart php8.5-fpm` y `sudo systemctl restart apache2`

```
miadmin@mdc-used1:~$ sudo systemctl restart php8.5-fpm
miadmin@mdc-used1:~$ sudo systemctl restart apache2
```

Y ahora comprobamos el servicio PHP → `sudo systemctl status php*-fpm`

```
miadmin@mdc-used1:~$ sudo systemctl status php8.5-fpm
● php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php8.5-fpm.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 08:42:50 UTC; 1min 2s ago
     Invocation: d90ba43ff3384f65a272e20488da54f8
       Docs: man:php-fpm8.5(8)
    Process: 15442 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.5/fpm/pool.d/www.conf 85 (code=exited, status=0/SUCCESS)
   Main PID: 15439 (php-fpm8.5)
      Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0.00req/sec"
     Tasks: 3 (limit: 1891)
    Memory: 8M (peak: 9.6M)
       CPU: 84ms
    CGroup: /system.slice/php8.5-fpm.service
            └─15439 "php-fpm: master process (/etc/php/8.5/fpm/php-fpm.conf)"
              └─15440 "php-fpm: pool www"
                └─15441 "php-fpm: pool www"

oct 05 08:42:49 mdc-used1 systemd[1]: Starting php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager...
oct 05 08:42:50 mdc-used1 systemd[1]: Started php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager.
miadmin@mdc-used1:~$
```